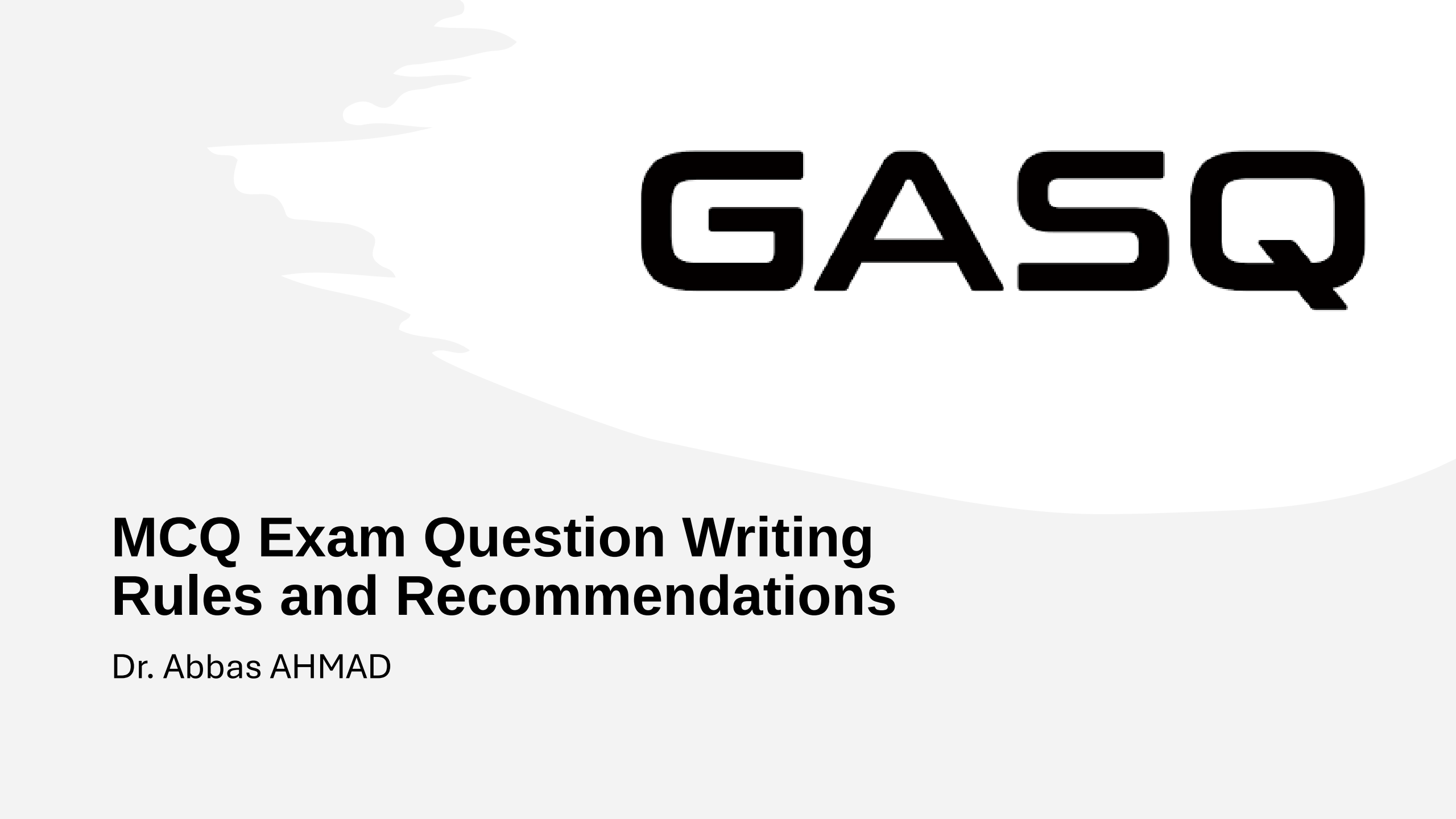


A light gray silhouette of the map of Pakistan is positioned in the upper left corner of the slide, serving as a background element.

GASQ

MCQ Exam Question Writing Rules and Recommendations

Dr. Abbas AHMAD

INTRODUCTION

- This presentation contains the rules for creating multiple choice questions for examinations as well as corresponding sample questions.
- In addition, it includes recommendations on best practices for question creation.

GLOSSARY

- **K-level** : Knowledge level, Cognitive level
- **LO** : Learning Objective
- **MC** : Multiple Choice

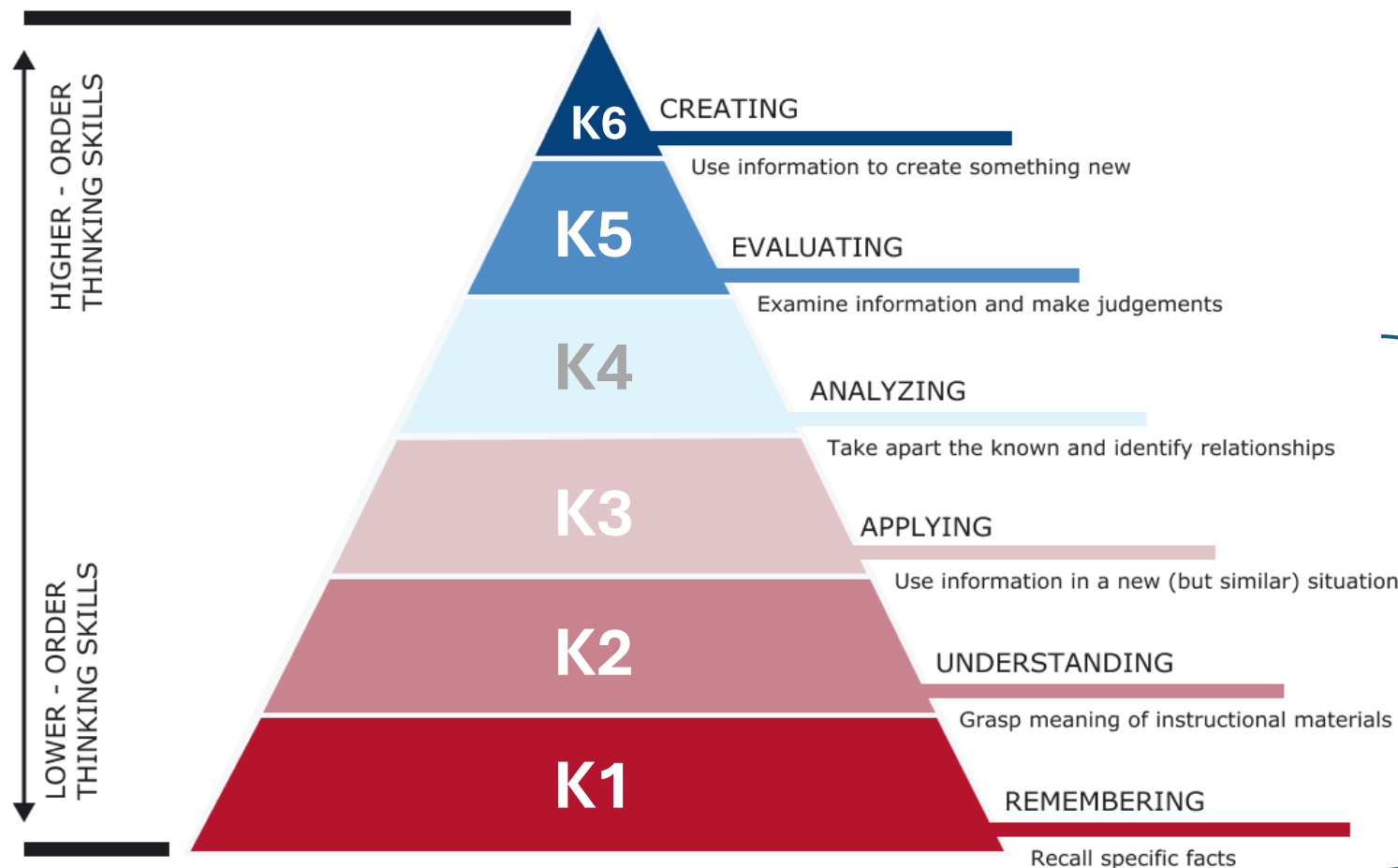
- **Exam question** : An individual component of the examination which is used to assess the knowledge of a candidate's understanding of the syllabus material and achievement of Learning Objectives.
 - Synonym : Exam item
- **Examination / Exam** : An assessment intended to measure a candidate's (test-taker's) knowledge, skill, and competence related to a syllabus.
 - ISO 17024: Mechanism that is part of the evaluation, which measures a candidate's competence by one or more means such as written, oral, practical and observational.

GLOSSARY

- **Learning Objective (LO)** : An outcome statement that captures specifically what knowledge, skills and behaviors a learner should be able to exhibit following instruction. Learning Objectives should:
 - Support the overall educational aims
 - Support the Business Outcomes defined for the syllabus
 - Drive the creation of examination questions
- **MC question (MC or MCQ)** : A type of exam question which consists of a problem and a list of suggested solutions (options). Single-answer, Matching and Multiple-Answer (Pick-Two) versions of this question type may be used in exams.
- **Scenario** : A detailed description of a situation which includes necessary background information for a series of examination questions
- **Distracter** : Incorrect or clearly inferior response option

Bloom's Taxonomy

BLOOM'S TAXOMY - COGNITIVE DOMAIN (2001)



Multiple choice questions are used to assess Knowledge levels K1 through K4.

Multiple Choice Questions : Basic concepts

- Seven different types of multiple choice questions:
 1. Type-A
 2. Roman
 3. Ordering (Out Of Scope)
 4. Matching Select
 5. Matching Drag (Out Of Scope)
 6. Grouping (Out Of Scope)
 7. Pick-Two

Multiple Choice Questions : Basic concepts

- A multiple choice question consists of two parts:
 - **Stem** – contains a question phrase and sufficient information to select the correct response option(s). A problem statement or a **scenario** may enhance the stem
 - **Response Options** – for Type-A, Roman, and Matching Select types of questions, there shall be one correct response option, while for Pick-Two questions there are two correct response options.
 - **Response Area** – for Ordering, Matching Drag and Grouping types of questions, there shall be a number of options within an response area, all options are first correct when “placed” correctly (Out Of Scope)

RULES

Rule-1 All K2-level through K6-level questions shall assess at least one Learning Objective

Rule-2 All K1-level questions shall assess at least one Learning Objective or test for recall of glossary terms

Rule-3 The ability to answer one question shall not be dependent on the answer from another question

Rule-4 Questions shall not require specialized domain-specific knowledge unless included in the syllabus

Rule-5 Questions shall not assess trivial or obscure knowledge

RULES

- **Rule-6** Questions shall use good grammar, punctuation, and spelling
- **Rule-7** Questions shall be written using simple, clear and unambiguous language, taking care to measure the candidate's knowledge of the syllabus, and not his or her vocabulary
- **Rule-8** Wording that could reasonably be interpreted as racist or sexist, or which may have a cultural or religious bias shall be avoided
- **Rule-9** Capitalization (e.g. MUST, MOSTLY and BEST) shall be used to emphasize key words that, if missed, could cause the candidate to select an incorrect answer
- **Rule-10** 'Double negative' phrasing (e.g. 'not inappropriate') shall not be used in both the question stem and answer options

RULES

- **Rule-11** No clues shall be given that can lead to the correct answer
- **Rule-12** Contractions (e.g. can't, don't, isn't, you'll, you're) shall not be used in either question stems or response options
- **Rule-14** Do not use 'all of the above' or 'none of the above' as options
- **Rule-15** MC questions at the K4 Level must contain a problem statement and supporting background information. The problem statement may be part of a larger scenario
- **Rule-16** MC questions must adhere to the structure as written in the Structure for Multiple Choice Question Types (Following)

RECOMMENDATIONS

- **Recommendation-1** Where applicable, use terms defined in the Glossary
- **Recommendation-2** Questions should use realistic situations, where feasible given other constraints, such as the time available for answering a question
- **Recommendation-3** Minimize the difficulty of calculations
- **Recommendation-4** Avoid the use of local or national references that are not known worldwide
- **Recommendation-5** Limit the use of NOT in exam questions

RECOMMENDATIONS

- **Recommendation-6** When “NOT” is used, be sure to capitalize the negative term, such as NOT, for emphasis
- **Recommendation-7** The stem should be written using a question format rather than in a statement format. For instance, it is better to write “What does a tester do during Static testing?” than “During Static testing the tester does the following:”
- **Recommendation-8** Avoid using absolutes such as ‘all’, ‘none’, ‘always’, and ‘never’
- **Recommendation-9** Use distracters that are plausible but clearly incorrect or sufficiently inferior to the best option
- **Recommendation-10** Incorporate common candidate mistakes in distracters, whenever possible

RECOMMENDATIONS

- **Recommendation-11** Avoid giving hints to the correct option based on option length or wording
- **Recommendation-12** Keep sentences short for list entries
- **Recommendation-13** To avoid confusion between MC question types, add a short line of instruction as appropriate, at the bottom of each question. Examples: “Select ONE answer”, “Select TWO answers”, or “Order these items in sequence”
- **Recommendation-14** MC questions may adhere to the guidelines as written in the Guidelines for using Multiple Choice Question Types (Following)

TYPE-A (Single correct option)

- Type-A type questions include a stem and four response options, including a single correct option.
- The stem consists of a limited amount of information.
- Each of the four response option consist of an answer to the question in the stem.
- In order to receive credit for correctly answering a Type-A type question, the candidate must select the correct response option.

TYPE-A STRUCTURE

[Stem text]

[4 response options (“a”, “b”, “c”, and “d”), as listed:]

a) [Response option a]

b) [Response option b]

c) [Response option c]

d) [Response option d]

Select ONE answer. *[Example of Recommendation-7]*

TYPE-A ISTQB® EXAMPLE

Level: K1

What does a tester do during "Static testing"?

- a) Reviews requirements and compares them with the design
- b) Runs the tests on the exact same setup each time
- c) Executes test to check that the hardware has been set up correctly
- d) Runs the same tests multiple times, and checks that the results are statistically meaningful

Select ONE answer.

Correct answer: a

ROMAN TYPE

- Roman type questions are used to measure the candidate's ability to assert statements.
- Roman type questions include a stem and four response options, including a single correct option.
- The stem consists of a limited amount of information and five statements, each preceded by a **roman numeral**.
- Each of the four response option consist of two or three roman numerals, representing the TRUE statements.

ROMAN TYPE

- Each roman entry must appear in at least two response options and not in more than three response options. (By using this technique, we avoid takers being able to determine which option is correct, when they only know that a single statement has a TRUE value).
- In the response options only list TRUE roman entries. Note, logically, with four response options, two options will list two TRUE statements and the other two will list three TRUE statements.
- In order to receive credit for correctly answering a Roman type question, the candidate must select the correct response option containing the correct combination of TRUE Statements from the choices provided in the set of five statements.

ROMAN TYPE STRUCTURE

[Stem text]

[List of 5 stem statements (“i”, “ii”, “iii”, “iv”, and “v”), as listed:]

- i. [Statement i]
- ii. [Statement ii]
- iii. [Statement iii]
- iv. [Statement iv]
- v. [Statement v]

[4 response options (“a”, “b”, “c”, and “d”), as listed:]

- a) [Two or three of i, ii, ii, iv and v]
- b) [Two or three of i, ii, ii, iv and v]
- c) [Two or three of i, ii, ii, iv and v]
- d) [Two or three of i, ii, ii, iv and v]

Select ONE answer. [Example of Recommendation-7]

ROMAN TYPE GUIDELINES

- Pay extra attention to the response options, it is easy to give away the correct answer
- This type of question is well suited for complex questions of a high K-level

ROMAN TYPE ISTQB® EXAMPLE

Level: K2

Which of the following statements are TRUE regarding when regression testing normally should be performed?

- i. Every week
- ii. After the software has changed
- iii. When the environment has changed
- iv. After check-in of each code change
- v. At acceptance test level

- a) ii, and iii
- b) i, iii, and v
- c) i, and iv
- d) ii, iv, and v

Select ONE option.

Correct answer: a

MATCHING SELECT TYPE

- Matching type questions are used to measure the candidate's ability to identify the relationship or association between related items.
- This format provides a way for candidates to connect an item: letter, number, word, sentence or phrase in one list to a corresponding letter, number, word, sentence or phrase in a second list.
- Matching Select type questions include a stem and four response options, including a single correct option.

MATCHING SELECT TYPE

- The stem consists of a limited amount of information and two lists. Each list must contain four items, the first four represented by a number and the second four with a capital letter.
- Each of the four response option consist of 4 pairs. Each pair consist of an item from the first list paired with an item from the second list. The same letter/number combination must not be used more than two times.
- In order to receive credit for correctly answering a Matching Select type question, the candidate must select the correct response option containing the correct pairing of the two lists.

MATCHING SELECT TYPE STRUCTURE

[Stem text]

[List of 4 items ("1", "2", "3", and "4"), as listed:]

1. [Item 1]
2. [Item 2]
3. [Item 3]
4. [Item 4]

[List of 4 items ("A", "B", "C", and "D"), as listed:]

- A. [Item A]
- B. [Item B]
- C. [Item C]
- D. [Item D]

[4 response options ("a", "b", "c", and "d"), as listed:]

- a) [Four combinations of 1.-4. and A.-D.]
- b) [Four combinations of 1.-4. and A.-D.]
- c) [Four combinations of 1.-4. and A.-D.]
- d) [Four combinations of 1.-4. and A.-D.]

Select ONE answer. *[Example of Recommendation-7]*

MATCHING SELECT TYPE GUIDELINES

- Pay extra attention to the response options, it is easy to give away the correct answer
- This type of question is well suited for complex questions of a high K-level

MATCHING SELECT TYPE ISTQB® EXAMPLE

Level: K2

Which general testing principles are characterized by the description below?

1. Early testing
 2. Defect clustering
 3. Pesticide paradox
 4. Absence-of-errors fallacy
-
- A. Testing should start at the beginning of the project
 - B. Conformance to requirements and fitness for use
 - C. A small number of modules contain the majority of defects
 - D. Test cases must be regularly reviewed and revised
-
- a. 1C, 2D, 3B and 4A
 - b. 1A, 2C, 3D and 4B
 - c. 1B, 2A, 3C and 4D
 - d. 1D, 2B, 3A and 4C

Select ONE answer.

Correct answer: b

PICK-TWO TYPE

- Pick-Two type questions include a stem and **five** response options, including two correct options.
- The stem consists of a limited amount of information.
- Each of the five response option consist of an answer to the question in the stem.
- In order to receive credit for correctly answering a Pick-Two type question, the candidate must select the two correct response option.
- The question is scored as either right or wrong, i.e. both of the selected options are correct, then the candidate is awarded all the points assigned to the question, otherwise they are awarded none.

PICK-TWO TYPE STRUCTURE

[Stem text]

[5 response options (“a”, “b”, “c”, “d” and “e”), as listed:]

a) *[Response option a]*

b) *[Response option b]*

c) *[Response option c]*

d) *[Response option d]*

e) *[Response option e]*

Select TWO answer. *[Example of Recommendation-7]*

PICK-TWO TYPE GUIDELINES

- Because the candidate have to select both correct options this type of question may, by nature, result in a lower amount of correct answers when compared to a Type-A type question

PICK-TWO TYPE ISTQB® EXAMPLE

Level: K2

A software system is already successfully running in an embedded environment. It is the only software system in this environment. It is now being ported to a PC-based environment and will be operated along with several other applications.

You need to select test design techniques to support compatibility testing. Which two of the following test design techniques are MOST likely to be chosen from?

- a. Equivalence partitioning
- b. Boundary value analysis
- c. Decision table testing
- d. All pairs testing
- e. Use-case testing

Select TWO answers.

Correct answers: d, e

SCENARIOS

- A scenario is a detailed description of a problem or situation which includes sufficient background information to answer one or more exam questions.
- In general, a scenario is written to provide context for a group of questions, with the scenario appearing before those questions.
- Scenarios can be used to support any question type

SCENARIOS

- The structure of the scenario is described in the following Table. Each of the elements should be considered when constructing the scenario and its associated questions:

Elements of a Scenario	Description
Environment/Background Information	Description of the organizational environment (Organization, System...) Information on the economic situation (Competitors; Regulations...) Information on the people involved in the situation, including their roles or functions, knowledge, communication style and behavior Constraints such as schedule, time budget, methods, techniques Any other supplementary information that is required for a complete answer
Situation/Problem Statement	Describes the problem that needs to be solved, or the specific situation, which is to be considered for each task

SCENARIOS : RULES

- **Rule-1** Scenarios shall be used for K4 and higher questions, but may also support K2 and K3 questions, as well.
- **Rule-2** A scenario shall be used when more than one question is based on the same situation
- **Rule-3** A scenario shall include all the information needed to understand the tasks / questions associated with it, either directly or by inference.
- **Rule-4** Review of the scenario shall be done together with its questions to ensure that the scenario contains all necessary information

SCENARIOS : RECOMMENDATIONS

- A scenario may contain superfluous information to provide candidates with an opportunity to identify what is relevant
- A scenario should not exceed 500 words
- Be aware of screen size if a scenario is presented through an electronic examination tool
- It is recommended to repeat the whole scenario every time it is used, clearly marking new/old parts. This prevent page shuffling which can be specially cumbersome if the scenario is presented through an electronic examination tool
- Scenario-related questions should follow the scenario to which they refer

SCENARIOS : EXAMPLE

You received the following defect report from the developers stating that the anomaly described in this test report is not reproducible.

Application hangs up

2022-May-03 – John Doe – Rejected

The application hangs up after entering "Test input: \$ä" in the Name field on the new user creation screen. Tried to log off, log in with test_admin01 account, same issue. Tried with other test admin accounts, same issue. No error message received; log (see attached) contains fatal error notification. Based on the test case TC-1305, the application should accept the provided input and create the user. Please fix with high priority, this feature is related to REQ-0012, which is a critical new business requirement.

What critical information is MISSING from this test report that would have been useful for the developers?

- a) Expected result and actual result
- b) References and defect status
- c) Test environment and test item
- d) Priority and severity

Select ONE option.

CHECKLIST FOR WRITING A QUESTION ON A SPECIFIC LEARNING OBJECTIVE

1. Is the Question understandable?
2. Is the Question consistent with the LO?
3. Is the Question written to match the K-level (does it use the correct action words) of the LO?
4. Is there a Rationale for each Response options?
5. Does the Question assume knowledge that is not in the syllabus, body-of-knowledge (if it exist) or is considered common knowledge?
6. Does the Question cover more than one LO? (It is not necessarily a bad thing)
7. Is there an overlap with other Questions for the covered LOs?
8. Is the Question giving away the answer to another question?
9. Is the Question as difficult as the Structure and Rules table intent? (A 1 point question should be easier than a 2 point question of the same K-level)